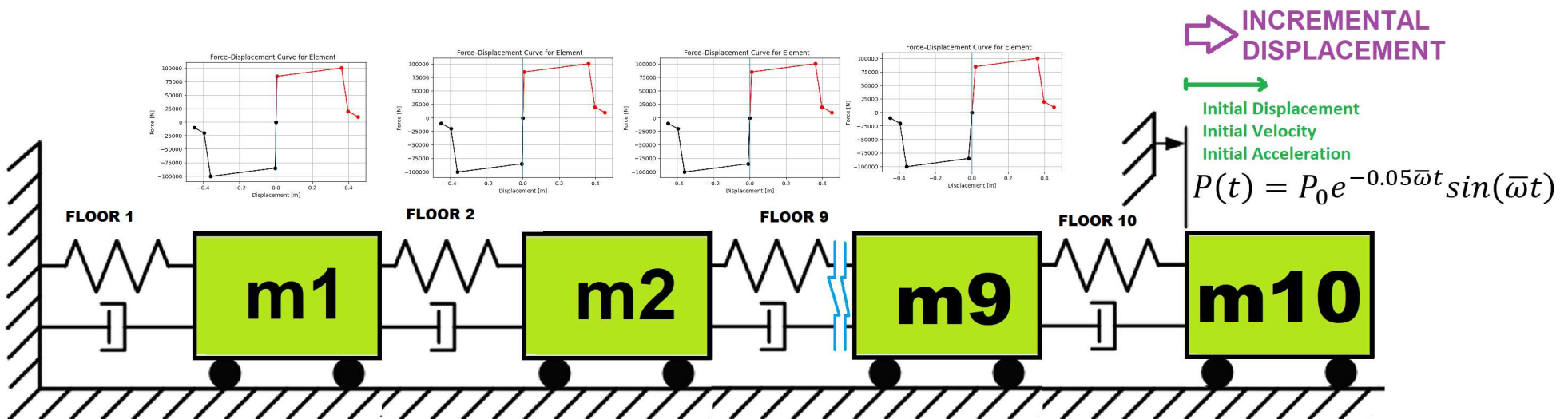


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

**FRAGILITY ANALYSIS BASED ON
ACCELERATION, STRUCTURAL DUCTILITY
DAMAGE INDEX, ENERGY DISSIPATION
CAPACITY INDEX, EQUIVALENT VISCOUS
DAMPING RATIO WITH NONLINEAR DYNAMIC
ANALYSIS OF A MULTI-DEGREE-OF-FREEDOM
(MDOF) SYSTEM UTILIZING 50 GROUND
MOTIONS IN OPENSEES**

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

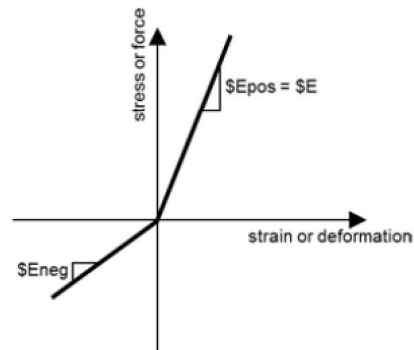
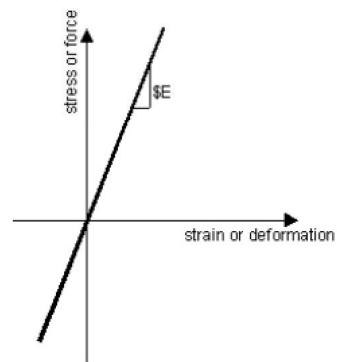
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

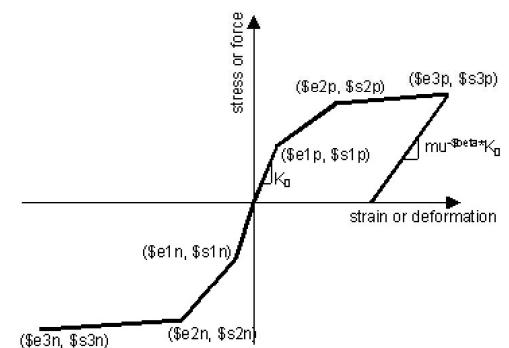
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

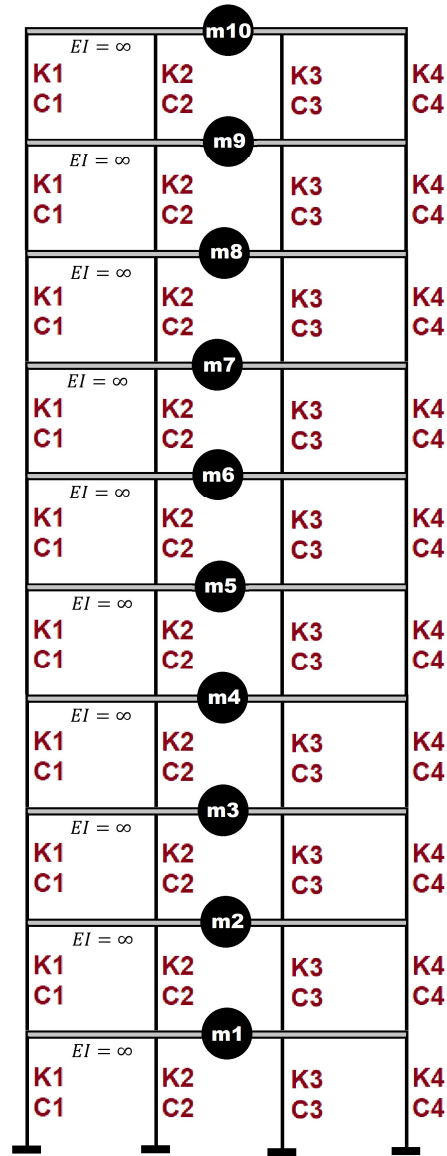
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



INELASTIC MATERIAL



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

Δ_u = Lateral Ultimate Displacement from Pushover Analysis

C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...UM_EDCI_EVDR+\MDOF_RESPONSE_SPECTRUM_EDCI_EVDR.py

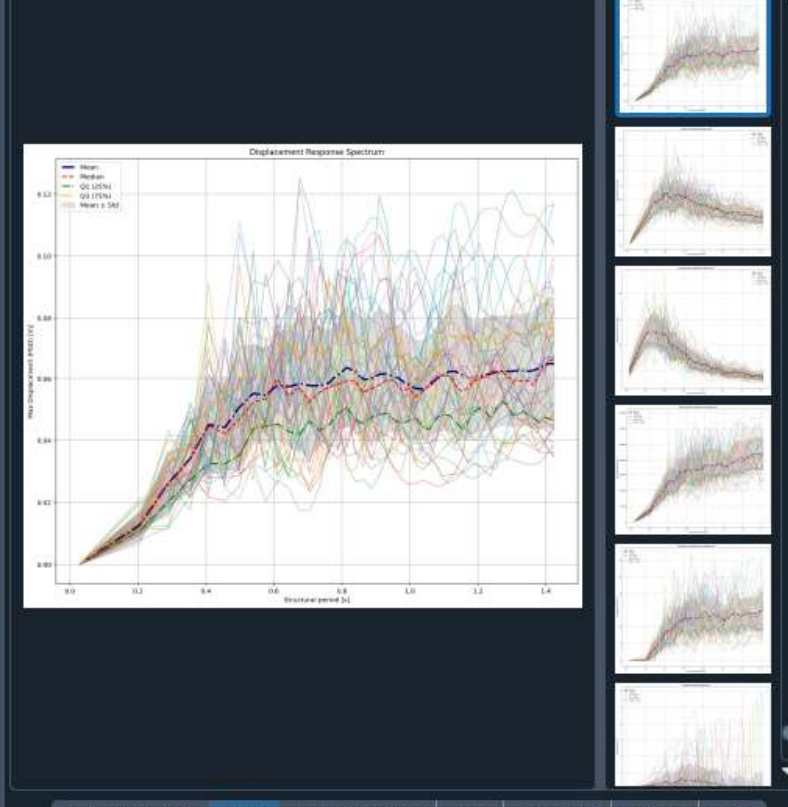
MDOF_RESPONSE_SPECTRUM_EDCI_EVDR.py ×
 OPENSEES_HYSTERETICSM_FORCE_DISP_FUN.py ×
 EQUIVALENT_VISCOUS_

```

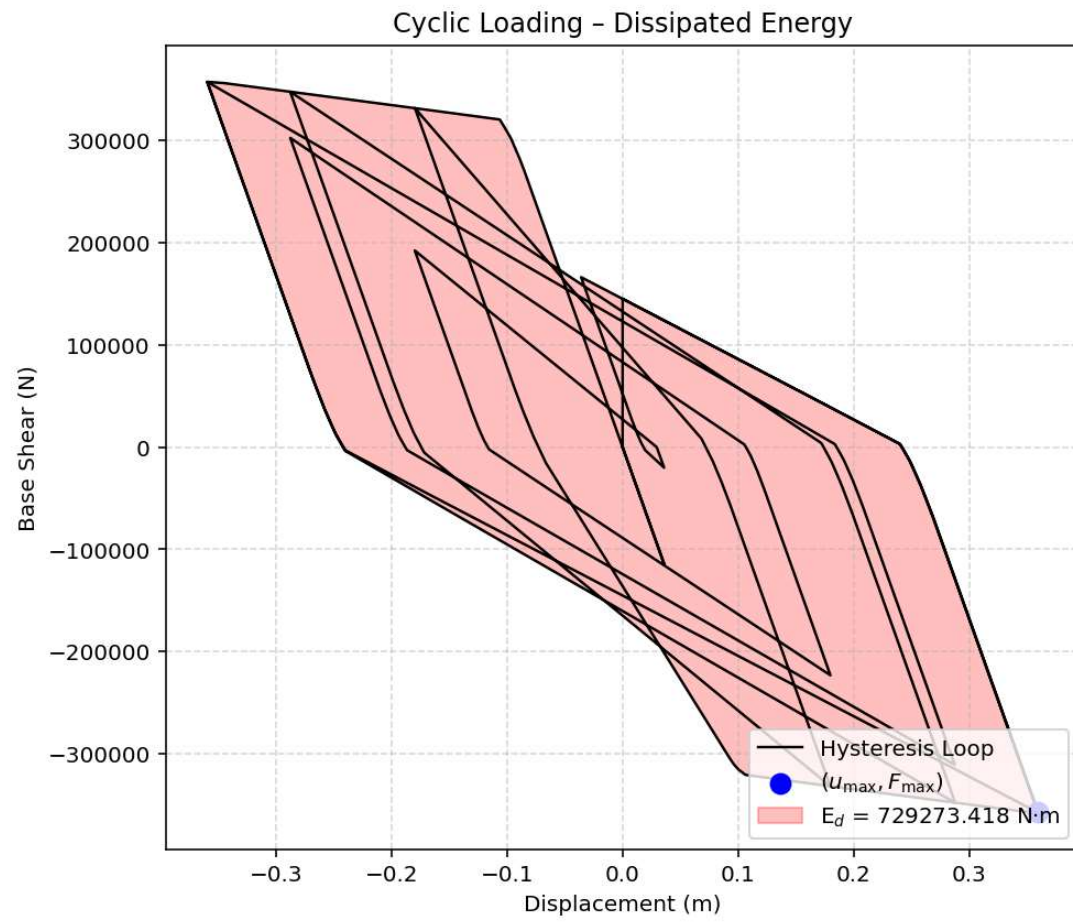
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # FRAGILITY ANALYSIS BASED ON ACCELERATION, STRUCTURAL DUCTILITY DAMAGE INDEX, ENERGY DISSIPATION
4 # CAPACITY INDEX, EQUIVALENT VISCOUS DAMPING RATIO WITH NONLINEAR DYNAMIC ANALYSIS
5 # OF A MULTI-DEGREE-OF-FREEDOM (MDOF) SYSTEM UTILIZING 50 GROUND MOTIONS IN OPENSEES
6 #-----
7 # INELASTIC RESPONSE SPECTRUM ANALYSIS
8 #-----
9 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
10 #-----
11 # ENERGY DISSIPATION CAPACITY INDEX =  $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacement})$ 
12 #-----
13 # This program performs Dynamic Analysis on a Multi-Degree-of-Freedom (MDOF) system
14 # subjected to 50 seismic ground motions. The analysis evaluates the structural response under varying
15 # levels of seismic intensity.
16 # The framework is designed to support researchers and engineers in assessing the probabilistic seismic
17 # performance of structures, with a focus on understanding the impact of uncertainty on structural
18 # response and design.
19 #-----
20 # Key Features:
21 # - Simulation of MDOF system using OpenSees.
22 # - Incremental scaling of ground motions for Nonlinear Dynamic Analysis.
23 # - Probabilistic fragility assessment based on predefined damage states.
24 # - Visualization of structural response and fragility curves.
25 # - Export of results for further analysis.
26 #-----
27 # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)
28 # EMAIL: salar.d.ghashghaei@gmail.com
29 #####
30 """
31
32 This code implements a comprehensive nonlinear dynamic analysis framework for
33 performance-based earthquake engineering assessment of multi-degree-of-freedom
34 (MDOF) systems. The methodology combines traditional nonlinear time-history
    
```

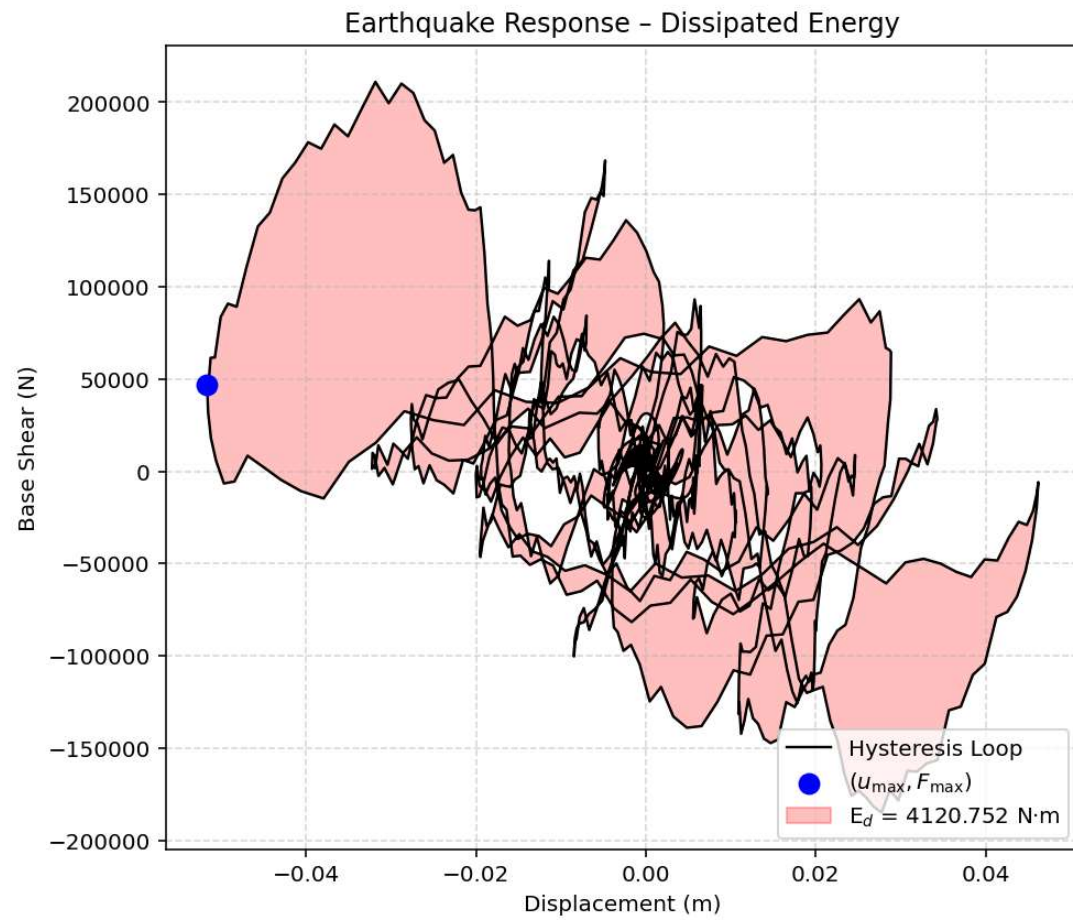
...OORS_04_COLUMNS_8_ANA\MDOF_RESPONSE_SPECTRUM_EDCI_EVDR+

21 %

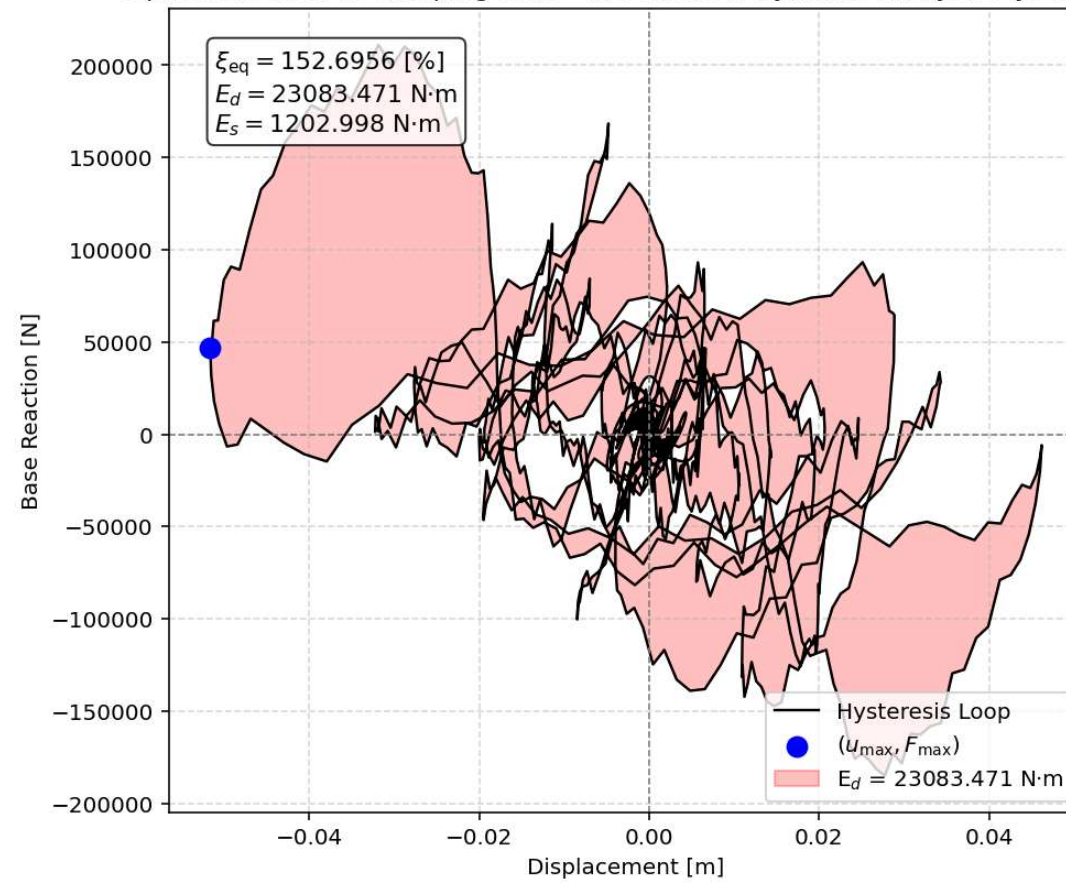


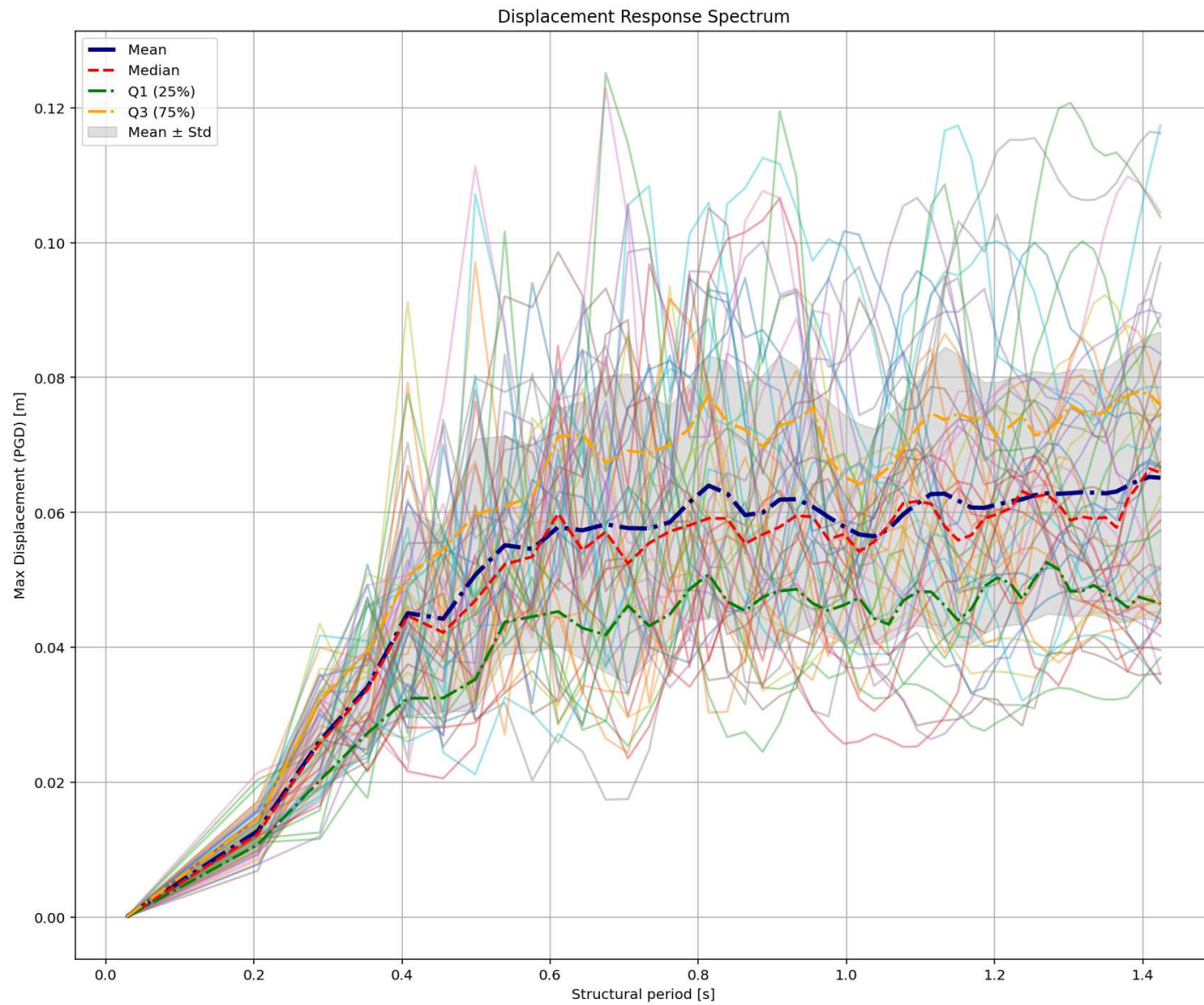
IPython Console Plots Variable Explorer Help Debugger History Files

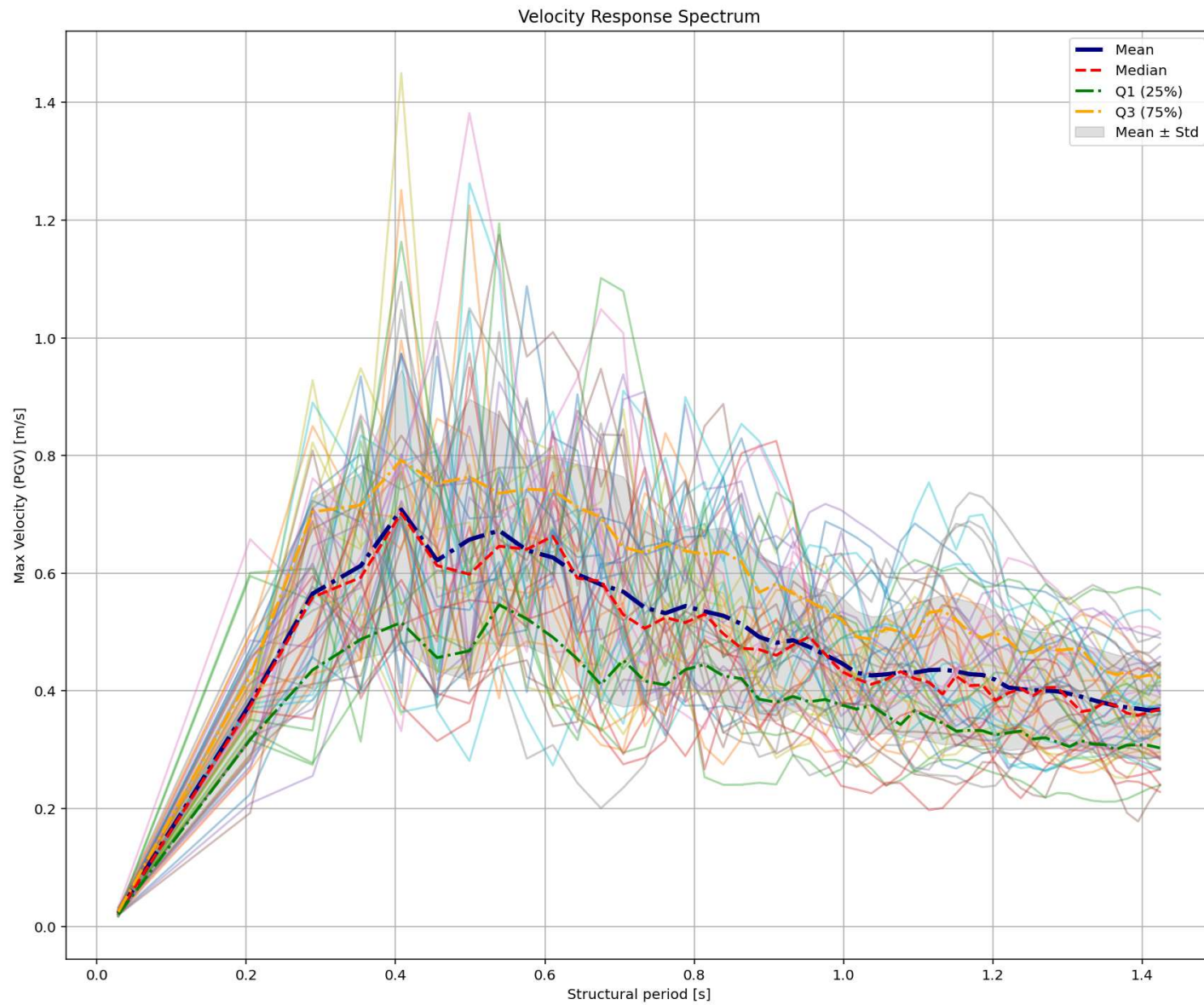


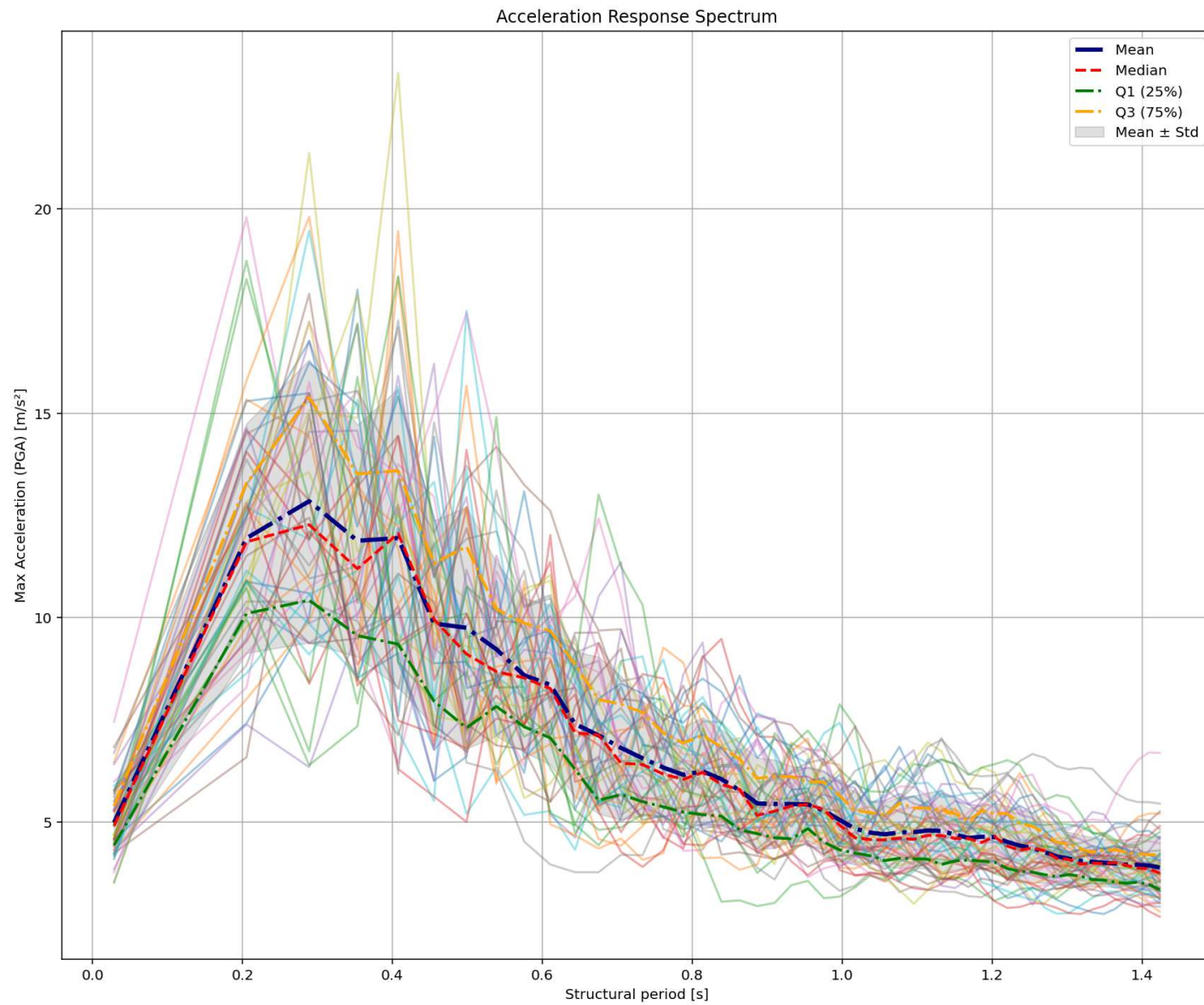


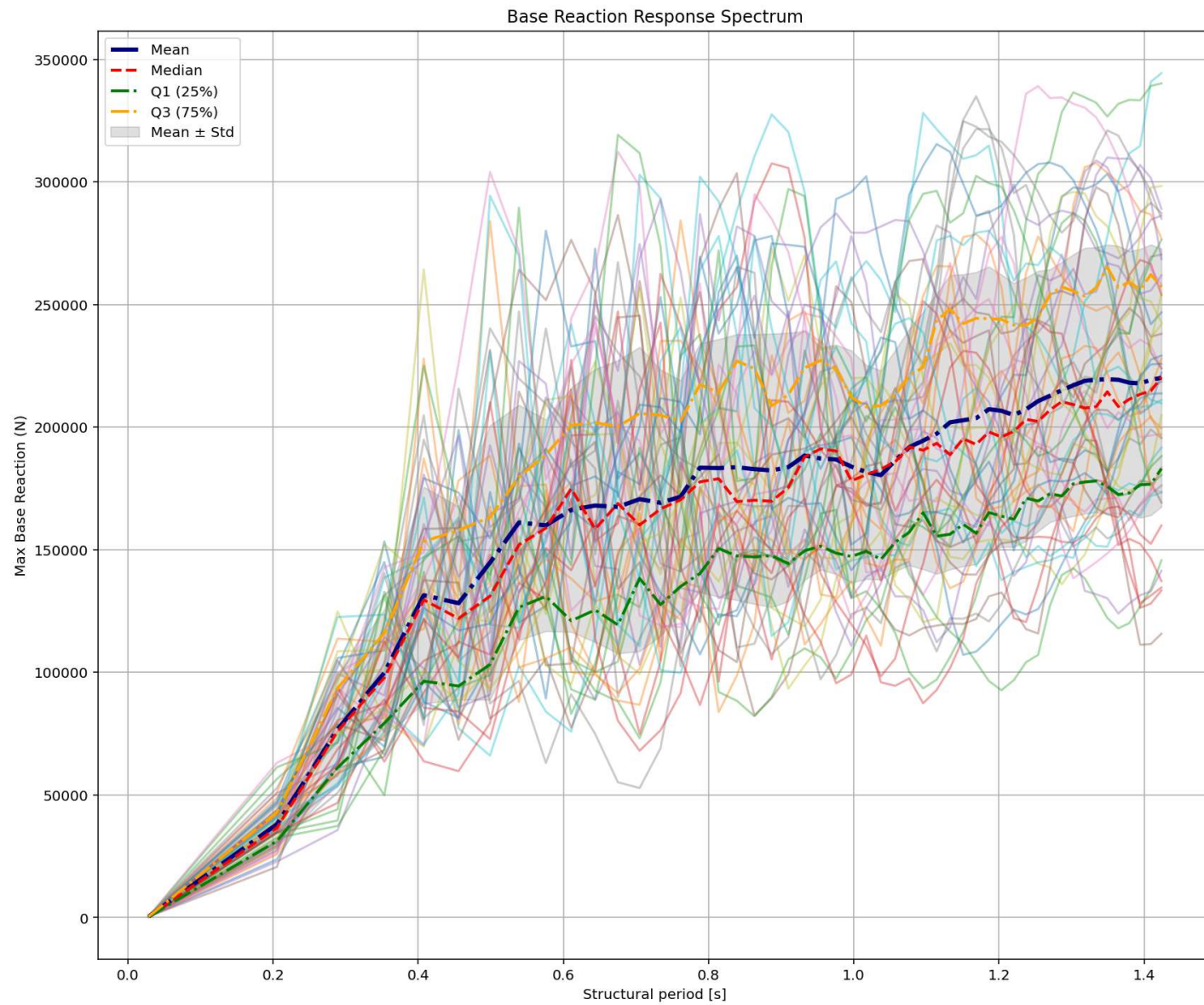
Equivalent viscous damping ratio - Incremental Dynamic Analysis Hysteresis

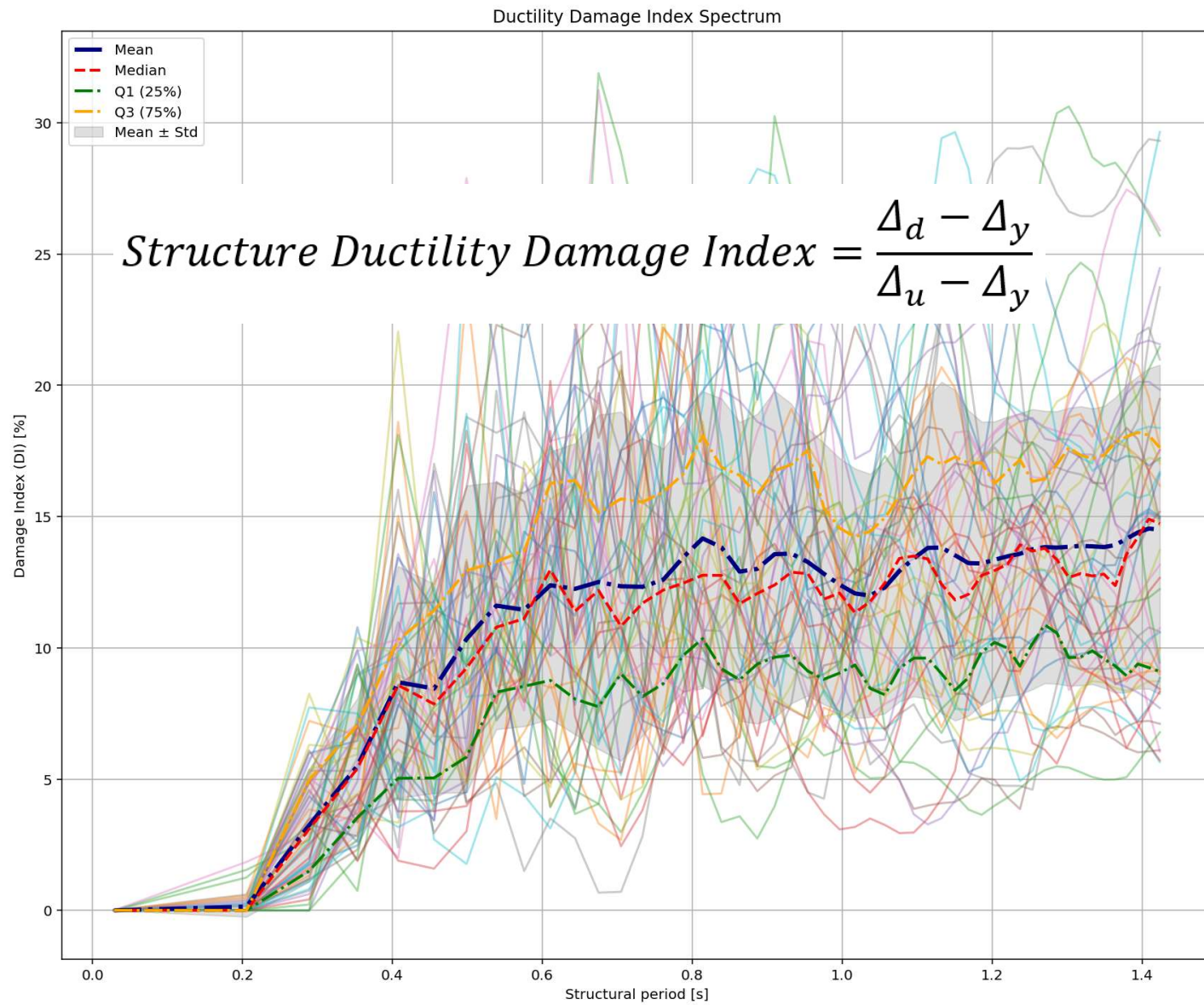


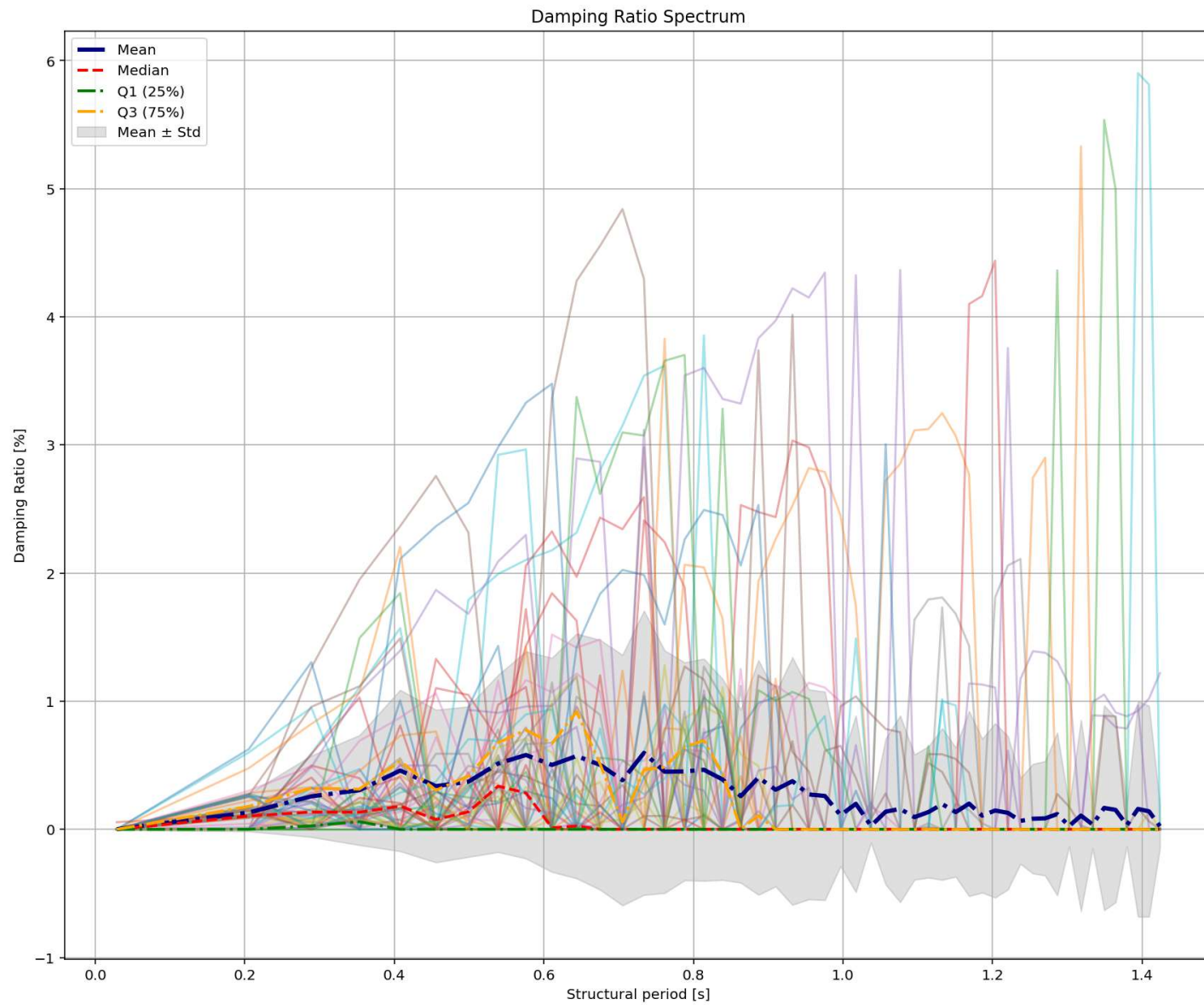


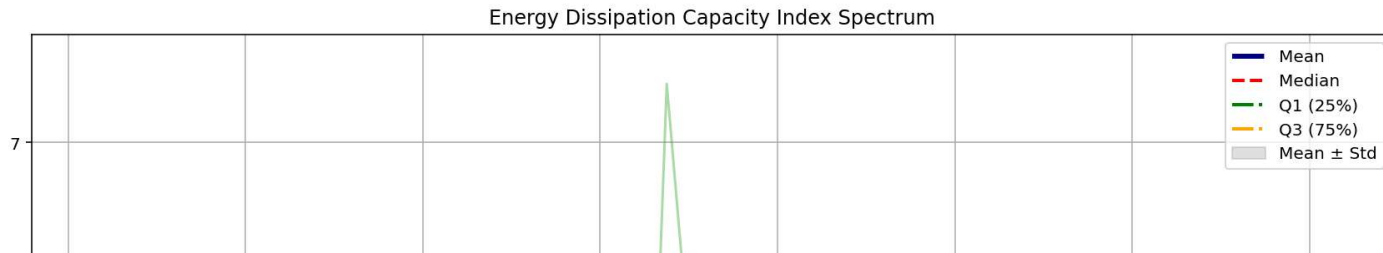




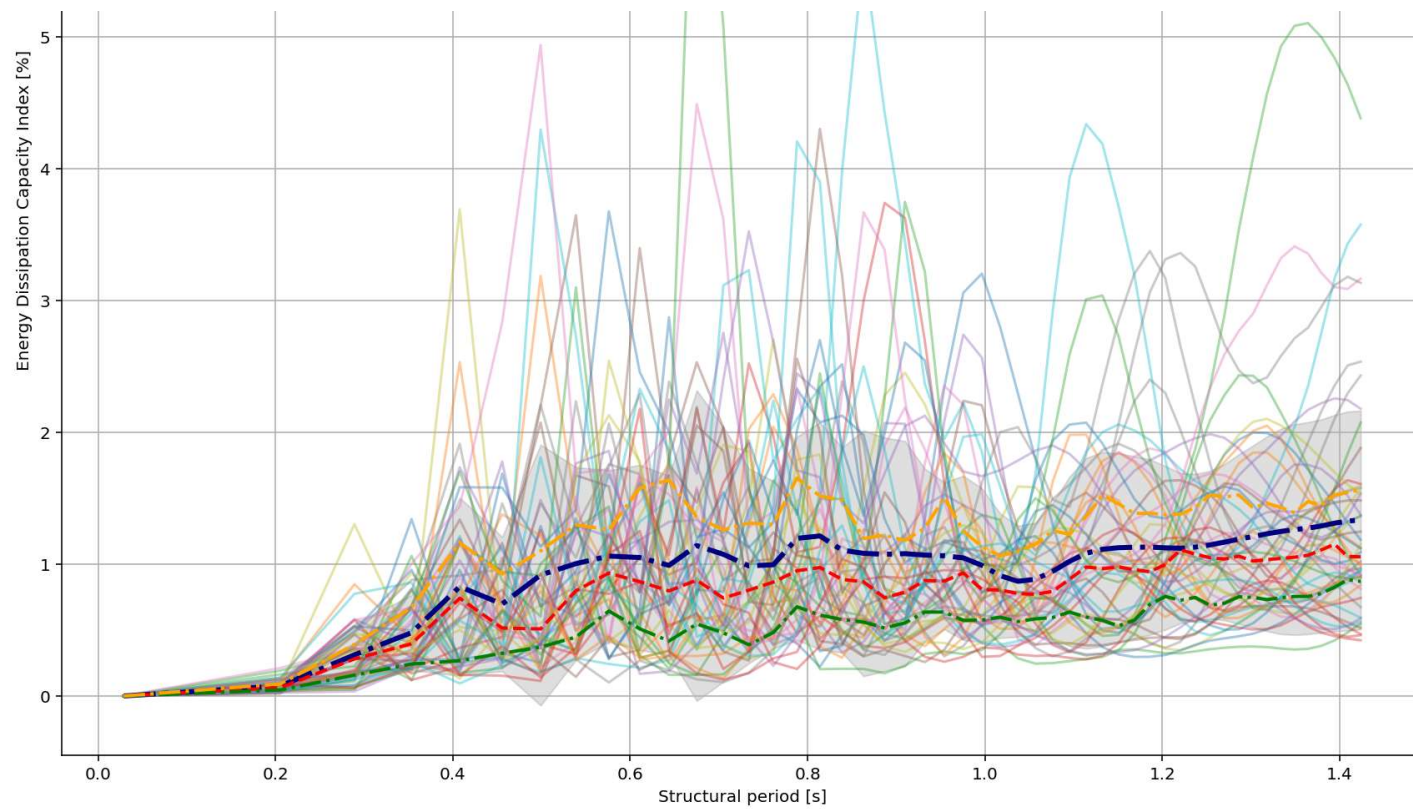




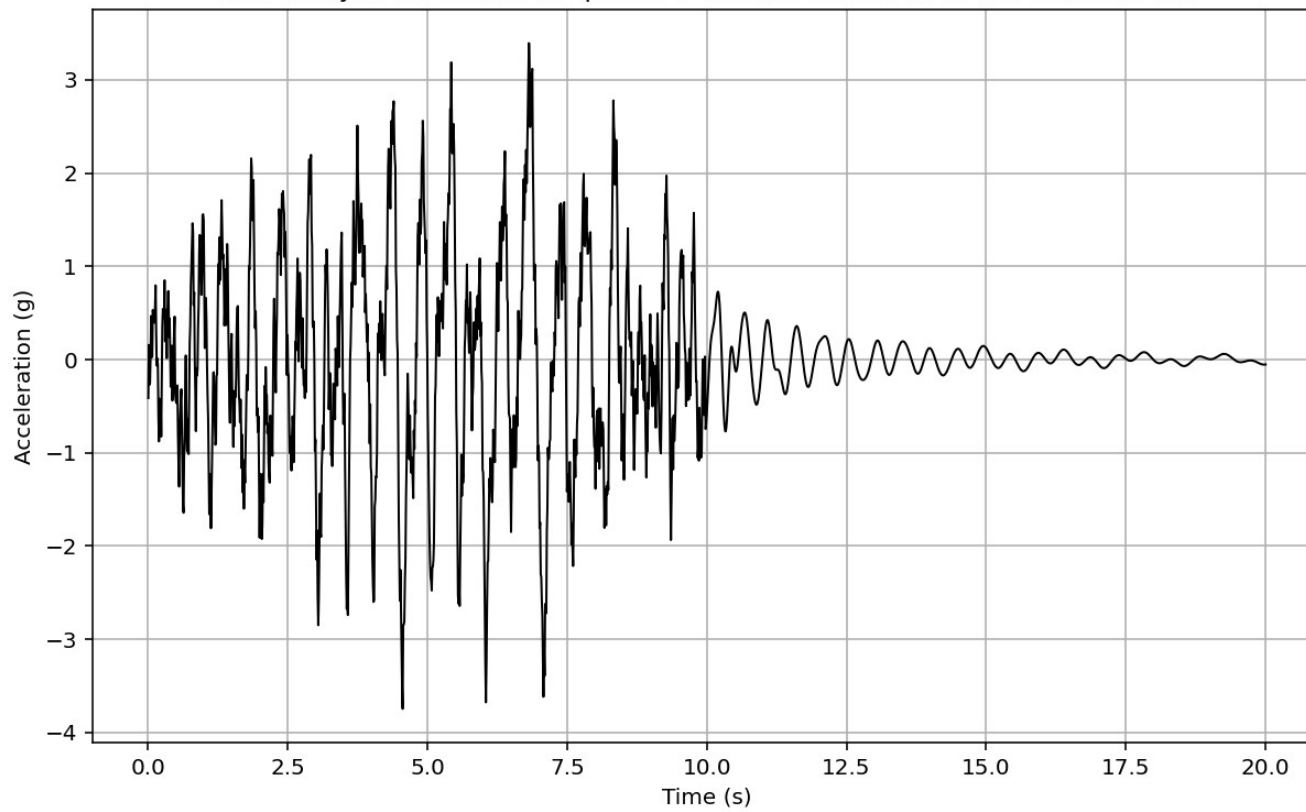


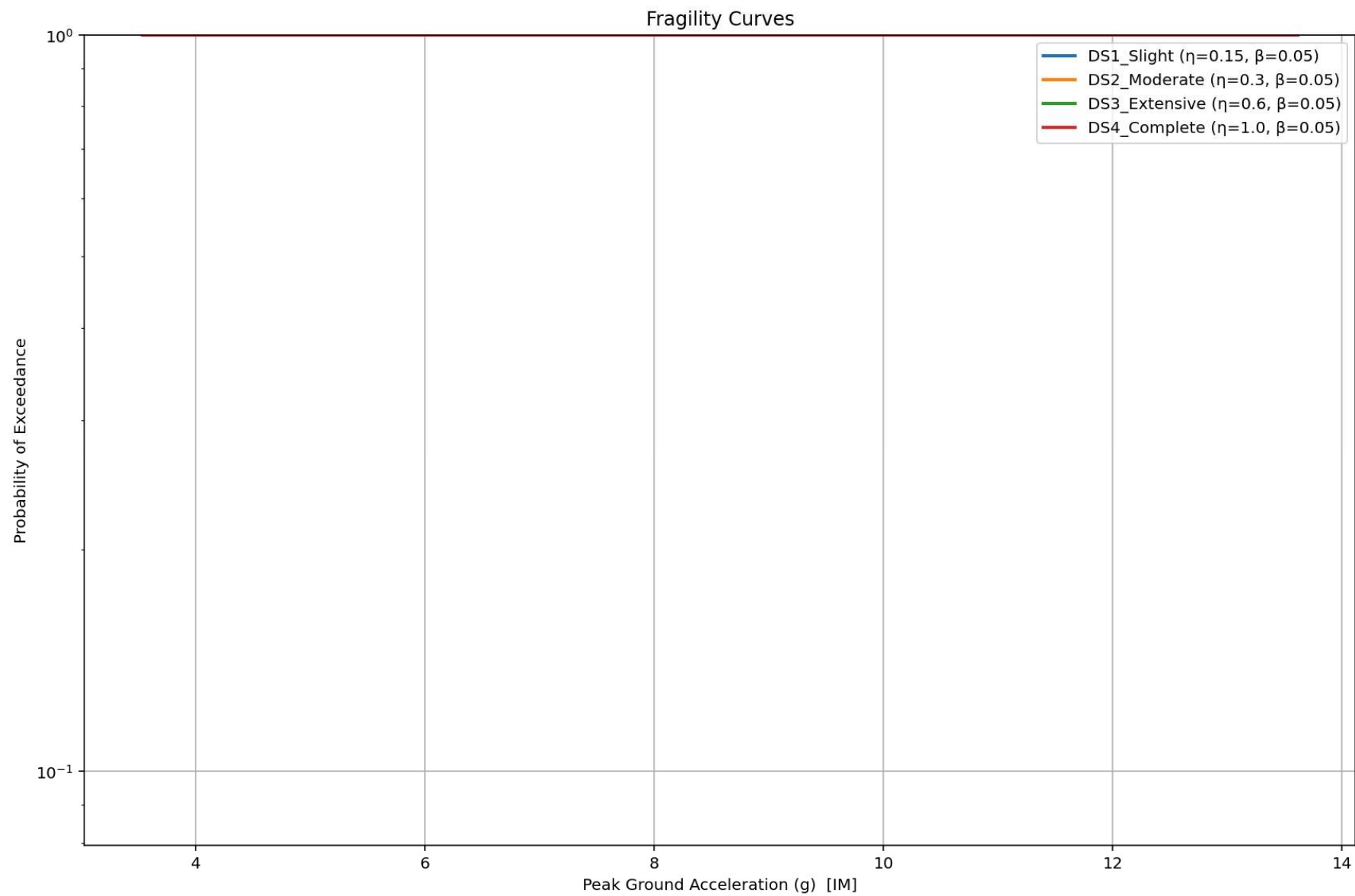


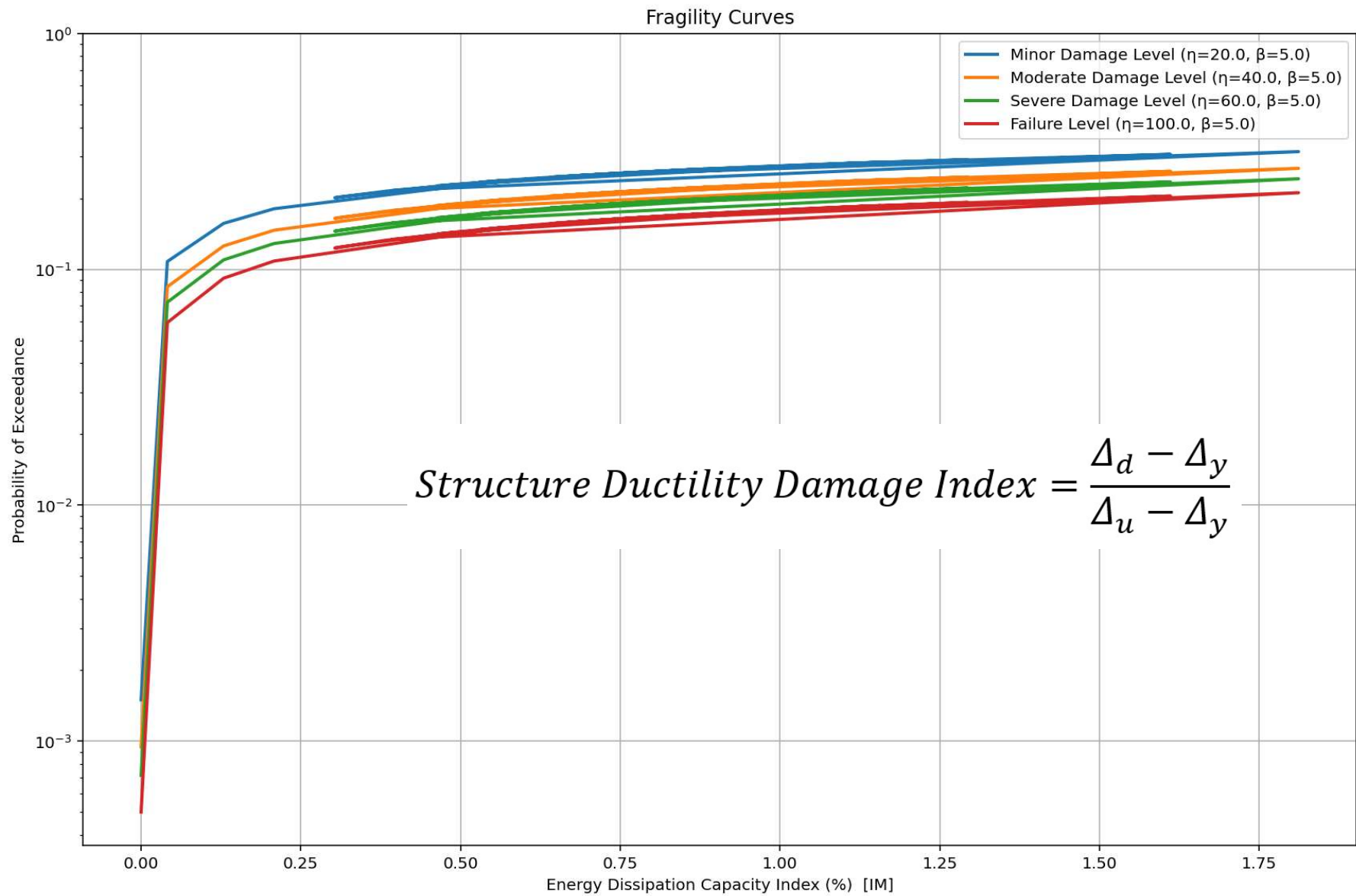
$$\text{Energy Dissipation Capacity Index} = \frac{E_d(\text{Dissipated Energy from Earthquake})}{E_d(\text{Dissipated Energy from Cyclic Displacement})} * 100$$

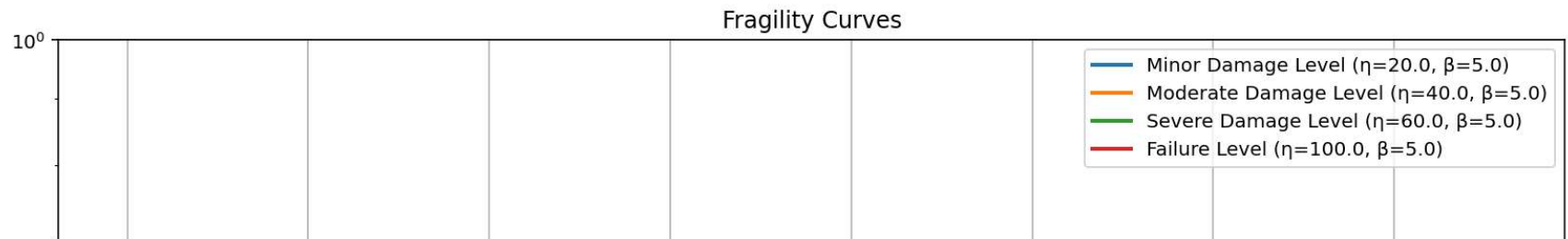


Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 3.7460

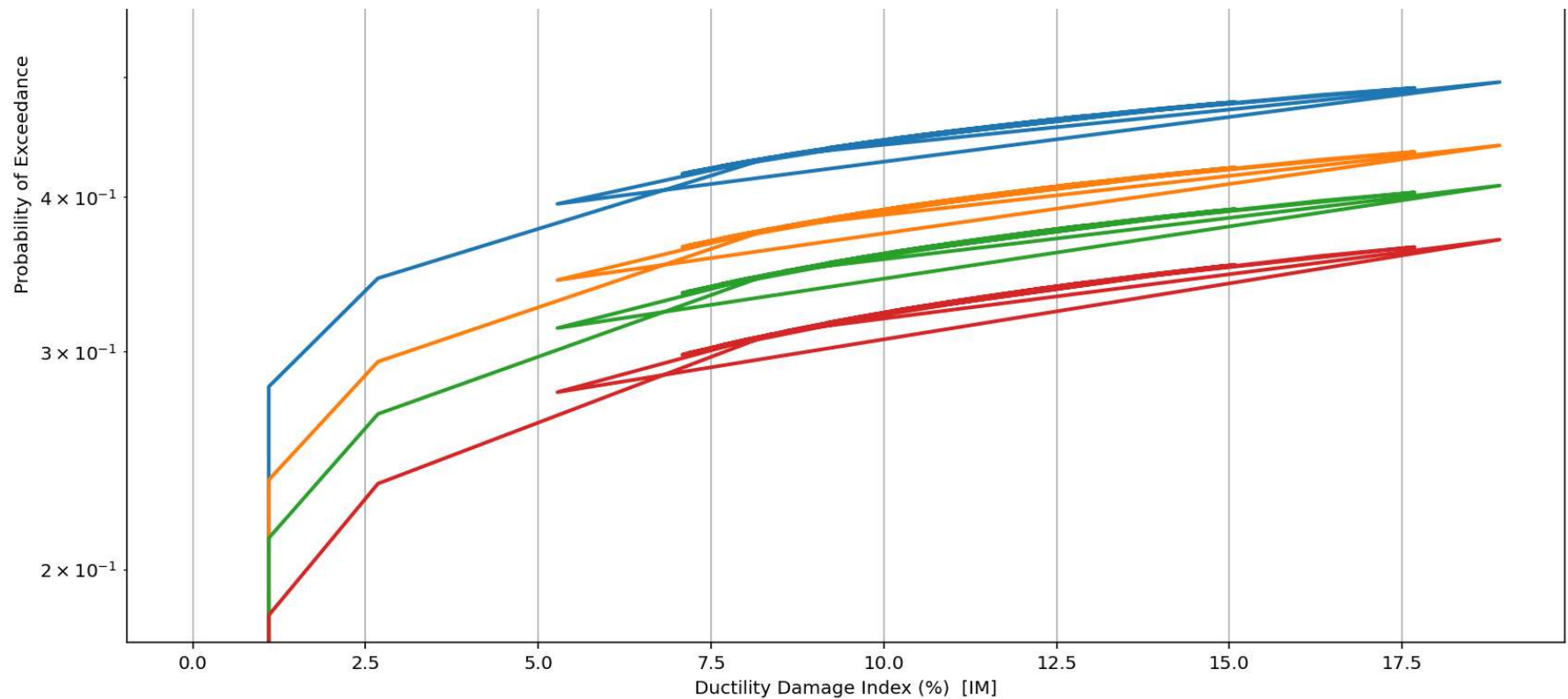


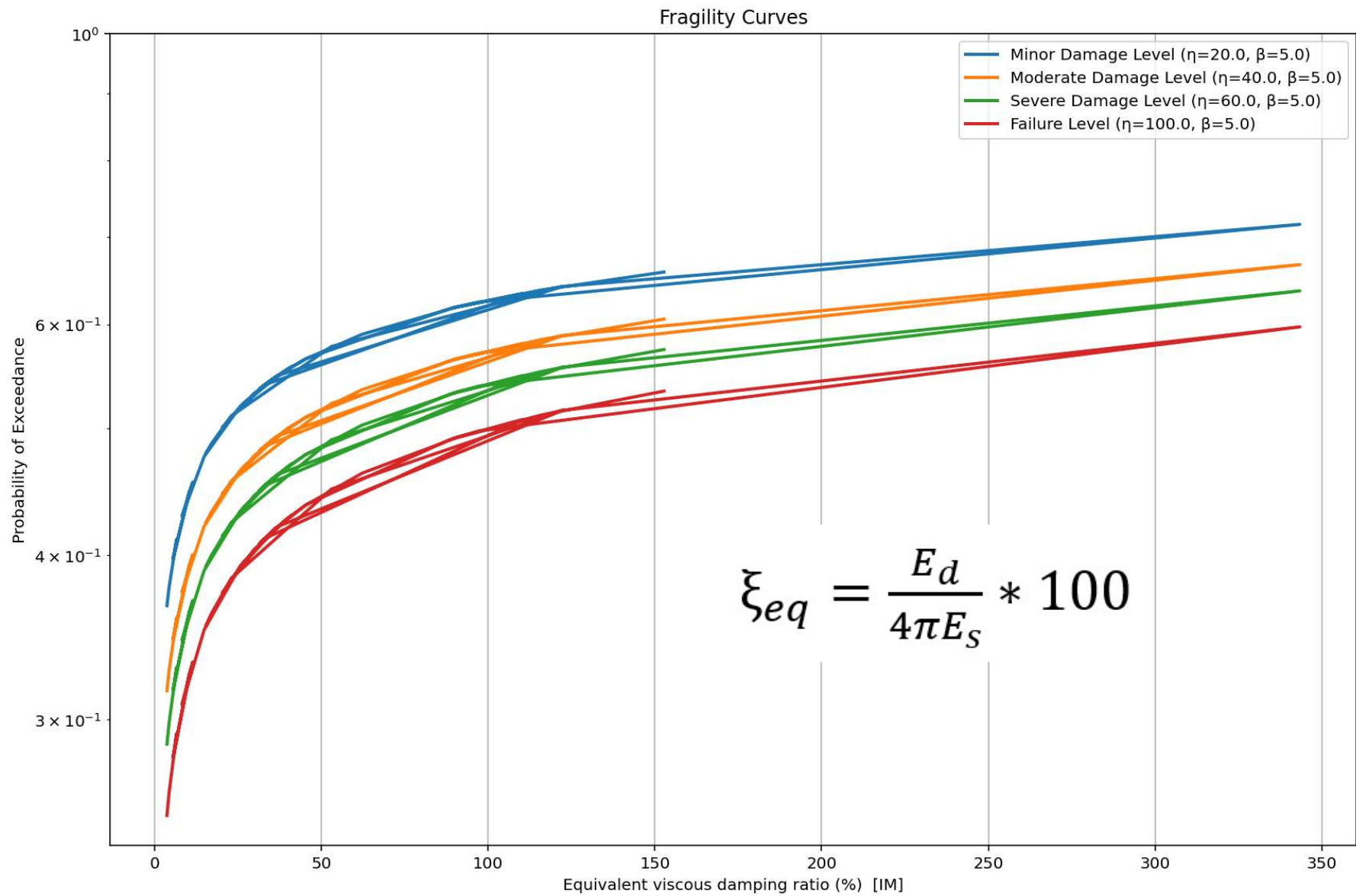




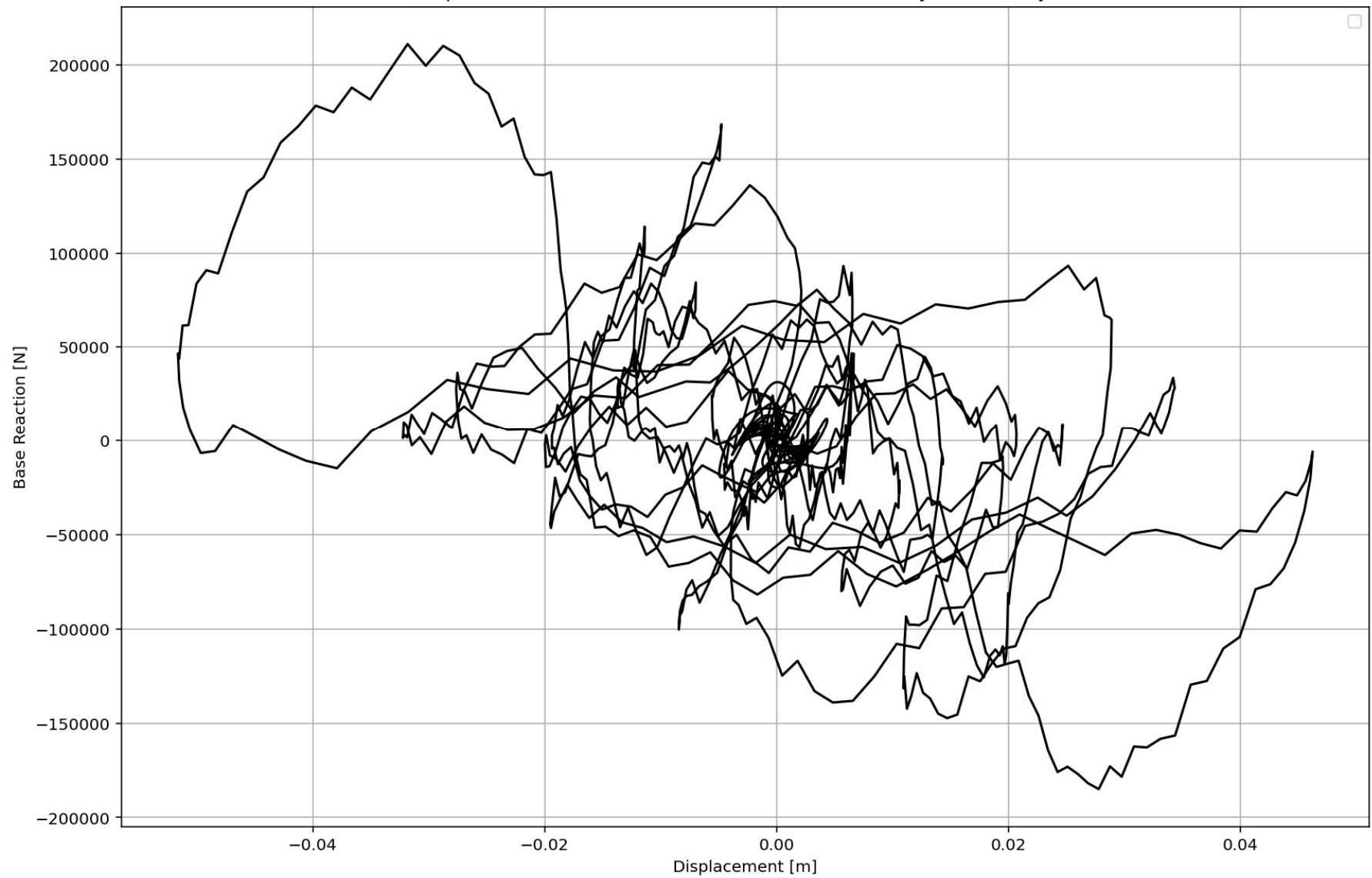


$$\text{Energy Dissipation Capacity Index} = \frac{E_d(\text{Dissipated Energy from Earthquake})}{E_d(\text{Dissipated Energy from Cyclic Displacement})} * 100$$





Displacement & Base Reaction Relation From Last Dynamic Analysis



Time vs Displacement - MAX. ABS: 0.05162845931289423

